KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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SCHOOL OF ELECTRONICS

PROJECT REPORT

on

VEHICLE MANAGEMENT SYSTEM USING OBJECT ORIENTED PROGRAMMING IN PYTHON

Submitted by:

Name : Ayushi Rai
Roll Number : 2430262
Course : B.Tech (ECS)
Department : ECS

Submitted to:
Faculty Name :
Professor Mayank Mishra

📌 PROBLEM STATEMENT

In real life, managing vehicle information such as cars and bikes can be difficult if done manually. There is a need for a simple software system that can manage different types of vehicles and store their details efficiently.

A system is required that can:

- Manage different types of vehicles
- Store vehicle details
- Allow comparison between vehicles
- Be simple and menu-driven for easy use

Traditional systems do not effectively demonstrate object-oriented programming concepts such as **inheritance and polymorphism**. Therefore, this project implements a Vehicle Management System using **Python OOP concept**

🎯 OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To develop a **menu-driven Vehicle Management System using Python**
- To implement **inheritance using a parent Vehicle class**
- To implement **polymorphism by handling Car and Bike objects together**
- To allow users to:
    - Configure cars and bikes
    - Save vehicle history
    - Compare two vehicles of the same type
- To understand real-world applications of **Python Object-Oriented Programming concepts**

🧠 CONCEPTS USED IN PYTHON

This project uses the following Python concepts:

- **Inheritance** – Car and Bike inherit from Vehicle
- **Abstraction** – Vehicle works as a base class
- **Polymorphism** – Vehicle reference stores Car and Bike objects
- **Encapsulation** – Data members are protected/private
- **List** – Used to store vehicles dynamically
- **Input Function** – Used for user input
- **Menu-Driven Programming**

🏗️ SYSTEM DESIGN (OVERVIEW)

```
   Vehicle (Base Class)
     |
  -----------------
 |          |
 Car         Bike
       |
VehicleManagement (Main Program)
```

- `Vehicle` → Abstract parent class
- `Car` → Child class
- `Bike` → Child class
- `VehicleManagement` → Main class (menu & control)

💻 PYTHON SOURCE CODE

```
class Vehicle:
    def __init__(self):
        self.type = ""
        self.company = ""
        self.model = ""
        self.fuel = ""

    def configure(self):
        pass

    def display(self):
        pass



class Car(Vehicle):

    def __init__(self):
        super().__init__()
        self.type = "Car"
        self.range = ""
        self.seats = ""
        self.sunroof = ""

    def configure(self):
```

```
        print("\nChoose Car Company:")
        print("1. Maruti\n2. Hyundai\n3. Tata")
        c = int(input())
        self.company = "Maruti" if c == 1 else "Hyundai" if c == 2 else "Tata"

        print("Choose Model:")
        print("1. Base\n2. Mid\n3. Top")
        m = int(input())
        self.model = "Base" if m == 1 else "Mid" if m == 2 else "Top"

        print("Choose Price Range:")
        print("1. 5-10 Lakh\n2. 10-15 Lakh\n3. 15-20 Lakh")
        r = int(input())
        self.range = "5-10 Lakh" if r == 1 else "10-15 Lakh" if r == 2 else "15-20
Lakh"

        print("Choose Seats:")
        print("1. 4\n2. 5")
        s = int(input())
        self.seats = "4" if s == 1 else "5"

        print("Choose Fuel:")
        print("1. Petrol\n2. Diesel\n3. Electric")
        f = int(input())
        self.fuel = "Petrol" if f == 1 else "Diesel" if f == 2 else "Electric"

        print("Sunroof:")
        print("1. Yes\n2. No")
        sr = int(input())
        self.sunroof = "Yes" if sr == 1 else "No"

        print("Car saved successfully!\n")

    def display(self):
        print("Type     :", self.type)
        print("Company  :", self.company)
        print("Model    :", self.model)
        print("Range    :", self.range)
        print("Seats    :", self.seats)
        print("Fuel     :", self.fuel)
        print("Sunroof  :", self.sunroof)


# ================ BIKE CLASS ================
class Bike(Vehicle):
```

```python
    def __init__(self):
        super().__init__()
        self.type = "Bike"
        self.engine = ""
        self.abs = ""

    def configure(self):
        print("\nChoose Bike Company:")
        print("1. Hero\n2. Honda\n3. Yamaha")
        c = int(input())
        self.company = "Hero" if c == 1 else "Honda" if c == 2 else "Yamaha"

        print("Choose Model:")
        print("1. Standard\n2. Sports\n3. Premium")
        m = int(input())
        self.model = "Standard" if m == 1 else "Sports" if m == 2 else "Premium"

        print("Choose Engine CC:")
        print("1. 100-125cc\n2. 150-160cc\n3. 200+cc")
        e = int(input())
        self.engine = "100-125cc" if e == 1 else "150-160cc" if e == 2 else "200+cc"

        print("Choose Fuel:")
        print("1. Petrol\n2. Electric")
        f = int(input())
        self.fuel = "Petrol" if f == 1 else "Electric"

        print("ABS:")
        print("1. Yes\n2. No")
        a = int(input())
        self.abs = "Yes" if a == 1 else "No"

        print("Bike saved successfully!\n")

    def display(self):
        print("Type     :", self.type)
        print("Company  :", self.company)
        print("Model    :", self.model)
        print("Engine   :", self.engine)
        print("Fuel     :", self.fuel)
        print("ABS      :", self.abs)


vehicles = []

while True:
```

```python
print("\n==============================")
print(" VEHICLE MANAGEMENT SYSTEM")
print("==============================")
print("1. Configure Car")
print("2. Configure Bike")
print("3. Show Saved Vehicles")
print("4. Compare Vehicles")
print("5. Exit")

choice = int(input("Enter choice: "))

if choice == 1:
    car = Car()
    car.configure()
    vehicles.append(car)

elif choice == 2:
    bike = Bike()
    bike.configure()
    vehicles.append(bike)

elif choice == 3:
    print("\n====== SAVED VEHICLES ======")
    for i, v in enumerate(vehicles):
        print("\nIndex:", i)
        v.display()

elif choice == 4:
    i1 = int(input("Enter first vehicle index: "))
    i2 = int(input("Enter second vehicle index: "))

    if i1 >= len(vehicles) or i2 >= len(vehicles):
        print("Invalid index!")

    elif vehicles[i1].type != vehicles[i2].type:
        print("Cannot compare different vehicle types!")

    else:
        print("\n===== COMPARISON =====")
        vehicles[i1].display()
        print("---------------------")
        vehicles[i2].display()

elif choice == 5:
    print("Thank you for using Vehicle Management System.")
```

```
        break

    else:
        print("Invalid choice!")
```

📌 SAMPLE OUTPUT

Project: Vehicle Management System using Python.

▶️ PROGRAM START

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 1
```

🚗 CONFIGURE FIRST CAR

```
Choose Car Company:
1. Maruti
2. Hyundai
3. Tata
2
Choose Model:
1. Base
2. Mid
3. Top
2
Choose Price Range:
1. 5-10 Lakh
2. 10-15 Lakh
3. 15-20 Lakh
2
Choose Seats:
1. 4
2. 5
4
Choose Fuel:
1. Petrol
2. Diesel
3. Electric
3
Sunroof:
1. Yes
2. No
1
Car saved successfully!
```

🚗 CONFIGURE SECOND CAR

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 1

Choose Car Company:
1. Maruti
2. Hyundai
3. Tata
3
Choose Model:
1. Base
2. Mid
3. Top
3
Choose Price Range:
1. 5-10 Lakh
2. 10-15 Lakh
3. 15-20 Lakh
3
Choose Seats:
1. 4
2. 5
5
```

```
Choose Fuel:
1. Petrol
2. Diesel
3. Electric
2
Sunroof:
1. Yes
2. No
2
Car saved successfully!
```

⚖️ COMPARE TWO CARS

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 4
Enter first vehicle index: 0
Enter second vehicle index: 1

===== COMPARISON =====
Type     : Car
Company  : Hyundai
Model    : Mid
Range    : 10-15 Lakh
Seats    : 5
Fuel     : Electric
Sunroof  : Yes
---------------------
Type     : Car
Company  : Tata
Model    : Top
Range    : 15-20 Lakh
Seats    : 5
Fuel     : Diesel
Sunroof  : No
```

🏍️ CONFIGURE FIRST BIKE

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 2

Choose Bike Company:
1. Hero
2. Honda
3. Yamaha
1
Choose Model:
1. Standard
2. Sports
3. Premium
2
Choose Engine CC:
1. 100-125cc
2. 150-160cc
3. 200+cc
2
Choose Fuel:
1. Petrol
2. Electric
2
```

```
ABS:
1. Yes
2. No
1
Bike saved successfully!
```

🏍️ CONFIGURE SECOND BIKE

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 2

Choose Bike Company:
1. Hero
2. Honda
3. Yamaha
1
Choose Model:
1. Standard
2. Sports
3. Premium
3
Choose Engine CC:
1. 100-125cc
2. 150-160cc
3. 200+cc
3
Choose Fuel:
1. Petrol
2. Electric
1
```

```
ABS:
1. Yes
2. No
1
Bike saved successfully!
```

⚖️ COMPARE TWO BIKES

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 4
Enter first vehicle index: 2
Enter second vehicle index: 3

===== COMPARISON =====
Type      : Bike
Company   : Hero
Model     : Sports
Engine    : 150-160cc
Fuel      : Electric
ABS       : Yes
---------------------
Type      : Bike
Company   : Hero
Model     : Premium
Engine    : 200+cc
Fuel      : Petrol
ABS       : Yes
```

📋 SHOW ALL SAVED VEHICLES (HISTORY)

```
    ==============================
     VEHICLE MANAGEMENT SYSTEM
    ==============================
...
    1. Configure Car
    2. Configure Bike
    3. Show Saved Vehicles
    4. Compare Vehicles
    5. Exit
    Enter choice: 3

    ====== SAVED VEHICLES ======

    Index: 0
    Type      : Car
    Company   : Hyundai
    Model     : Mid
    Range     : 10-15 Lakh
    Seats     : 5
    Fuel      : Electric
    Sunroof   : Yes

    Index: 1
    Type      : Car
    Company   : Tata
    Model     : Top
    Range     : 15-20 Lakh
    Seats     : 5
    Fuel      : Diesel
    Sunroof   : No
```

```
    Sunroof   : No

    Index: 2
...
    Type      : Bike
    Company   : Hero
    Model     : Sports
    Engine    : 150-160cc
    Fuel      : Electric
    ABS       : Yes

    Index: 3
    Type      : Bike
    Company   : Hero
    Model     : Premium
    Engine    : 200+cc
    Fuel      : Petrol
    ABS       : Yes
```

EXIT PROGRAM

```
==============================
 VEHICLE MANAGEMENT SYSTEM
==============================
1. Configure Car
2. Configure Bike
3. Show Saved Vehicles
4. Compare Vehicles
5. Exit
Enter choice: 5
Thank you for using Vehicle Management System.
```

CONCLUSION

This project successfully demonstrates the use of **Python Object-Oriented Programming concepts** in a real-world application. By using **inheritance and polymorphism**, the Vehicle Management System efficiently manages different types of vehicles such as cars and bikes.

The system is **menu-driven, user-friendly, and flexible**, making it suitable for academic learning. It helps students understand how real-world applications can be developed using **Python OOP principles**.